The empirical recurrence for OEIS A264479, recovered from its tail

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Abstract

OEIS A264479 records "Empirical recurrence of order 84" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 4096$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 85$ — every index at which it can be written down. Its last coefficient is $c_{84} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A264479 is "Number of (5+1)X(n+1) arrays of permutations of 0..n*6+5 with each element having directed index change 0,1 1,0 2,1 or -1,-1.". It has offset 1 and begins

$$1, 16, 49, 456, 3540, 28489, 215607, 1711113, \dots$$

The entry states:

> Empirical recurrence of order 84 (see link above)

As of the "Last modified" line on the live entry (Nov 14 2015 15:40:04, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 4096$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 4096$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 8192$ exact terms from $n = 1$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{84} c_i\, a(n-i),$$

c_1	6	c_{22}	−233057068	c_{43}	628326825700	c_{64}	−1014783750
c_2	−1	c_{23}	416706186	c_{44}	−395115518257	c_{65}	622250587
c_3	185	c_{24}	2202780774	c_{45}	180079728224	c_{66}	−93275073
c_4	−535	c_{25}	1840272298	c_{46}	−566163740945	c_{67}	114885963
c_5	632	c_{26}	−2023360504	c_{47}	343765234695	c_{68}	−72992405
c_6	−6842	c_{27}	−6947621063	c_{48}	−147476072938	c_{69}	10746571
c_7	18690	c_{28}	−13574683637	c_{49}	398914123810	c_{70}	−8930037
c_8	−32254	c_{29}	11759868046	c_{50}	−234391099085	c_{71}	6041118
c_9	116091	c_{30}	11710232877	c_{51}	88490245655	c_{72}	−964932
c_{10}	−337966	c_{31}	60551649541	c_{52}	−218093086355	c_{73}	459031
c_{11}	739809	c_{32}	−46483184804	c_{53}	125488027731	c_{74}	−349259
c_{12}	−1192099	c_{33}	−4249325929	c_{54}	−39090640598	c_{75}	65611
c_{13}	3413869	c_{34}	−175661007541	c_{55}	90730028809	c_{76}	−14999
c_{14}	−8123572	c_{35}	125217224221	c_{56}	−52144329837	c_{77}	13903
c_{15}	6031319	c_{36}	−31761040507	c_{57}	12896239659	c_{78}	−3135
c_{16}	−21089059	c_{37}	358509032537	c_{58}	−28033042628	c_{79}	350
c_{17}	48640047	c_{38}	−242886448287	c_{59}	16362970278	c_{80}	−374
c_{18}	12521335	c_{39}	96351809271	c_{60}	−3239528130	c_{81}	91
c_{19}	75135381	c_{40}	−542563295185	c_{61}	6304060718	c_{82}	−6
c_{20}	−164141866	c_{41}	353370770763	c_{62}	−3765495246	c_{83}	5
c_{21}	−360459610	c_{42}	−159172486214	c_{63}	626313730	c_{84}	−1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 85$, and it is the recurrence OEIS A264479 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1 x^{84-1} - \cdots - c_{84}$. Its constant term is $-c_{84} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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